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Driving Safety Assessment via Deep Learning Models on CAN-Bus Sensor Streams

Nurbakyt Agybetov^a, Beibut Amirgaliyev^{a,*}

^a*Astana IT University, Astana, 010000, Kazakhstan*

Abstract

This paper proposes a deep learning framework for continuous driving safety assessment based on CAN-derived vehicle dynamics data. Despite recent advances in driver behavior analysis, existing approaches often rely on window-level evaluation protocols that may introduce information leakage and primarily focus on discrete classification, without providing continuous safety measures that reflect temporal variability in driving behavior. To address these limitations, a window-based approach is introduced, in which continuous sensor signals are segmented into overlapping temporal windows and analyzed using neural network models, including 1D convolutional neural networks (1D-CNN) and long short-term memory (LSTM) networks. A weighted safety scoring mechanism is further proposed to aggregate window-level predictions into a temporally consistent trip-level safety measure, allowing improved sensitivity to high-risk driving events. The proposed framework is evaluated on the UAH-DriveSet dataset using leakage-free 5-fold trip-level cross-validation. Experimental results demonstrate that the 1D-CNN achieves a window-level recall of 0.91 and an AUC of 0.94, while the LSTM attains a recall of 0.89 and an AUC of 0.93. In addition, cross-domain validation on a synthetic CAN-like dataset confirms the robustness and generalization capability of the proposed approach. The obtained results indicate that the proposed framework provides an effective and interpretable solution for real-time driving safety assessment using vehicle sensor data.

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1. Introduction

Modern vehicles often come with built-in computer systems along with several separate electronic controllers connected using a network like the CAN bus. Sensors run nonstop, gathering data including how fast the vehicle moves, how much force acts on it, or the twist created inside [1]. That information moves through the CAN bus, a hub where every digital component shares data. Something shows up - not only figures, yet shapes too: how sharp your stops are,

* Corresponding author. Tel.: +7-701-033-9111.

E-mail address: Beibut.amirgaliyev@astanait.edu.kz

when gears shift, maybe how your beat moves distinct across varied roads. These live logs run without pause, bridging holes between recollection by people behind wheels and real events unfolding. Yet each year, far too many are injured on roads globally, risky ways of driving remain key behind these numbers [2]. What used to be mere copies of crash records now faces tools shaped by rigid standards or visual watchers who miss the roots of danger. Lately, scientists have dug into car sensor outputs with sharper tools, uncovering patterns using live trail data analysis [3, 4, 19]. Yet most today fixate either on tiny snippets or giant aggregates, barely allowing immediate monitoring. A fresh take on the job centers measurable outcomes, pulling in data from CAN networks and onboard sensors gathered over set periods. This work adds several key elements. First, it presents a clear method for splitting CAN signal data into windows, allowing flexible overlap between segments. Instead of relying on standard tools, it compares two common network models - one using 1D convolutional layers, the other long short-term memory units - using trip-level cross-validation to avoid biased results. From each window's predictions, a running safety measure is calculated based on probability outputs across segments. Testing spans two domains: real-world driving logs made public by UAH, plus synthetic data mimicking actual vehicle networks. That breadth helps determine how well findings translate beyond single scenarios.

2. Related Work

In recent years, increasing attention has been devoted to the analysis of driving behavior using vehicle sensor data, particularly signals obtained from in-vehicle networks such as the Controller Area Network (CAN). These signals provide detailed information on vehicle dynamics, including speed, acceleration, and braking activity, allowing the characterization of driver behavior in real-world conditions [1, 18]. Fugiglando et al. (2018) demonstrated that meaningful driving behavior patterns can be extracted from a limited set of CAN signals collected in uncontrolled environments. Their work highlights the potential of vehicle sensor data to capture natural driving behavior without relying on external observation systems. More recently, deep learning approaches have been applied increasingly to time-series vehicle data. In particular, hybrid architectures combining convolutional and recurrent neural networks have been proposed to automatically extract temporal features from raw sensor streams, reducing the need for handcrafted features [6, 7]. These models have shown strong performance in detecting driving patterns associated with unsafe behavior, such as aggressive acceleration, harsh braking, and abrupt maneuvers. Telematics-based studies further confirm that such kinematic patterns are strongly correlated with unsafe driving and increased accident risk. Statistical analyses indicate that sudden braking, rapid acceleration, and high-speed driving are reliable indicators of hazardous behavior [5, 17]. Large-scale fleet data also supports these findings, demonstrating that irregular driving patterns align with elevated risk levels observed in real-world conditions. Despite these advances, several limitations remain. Many existing studies rely on window-level data partitioning, which may introduce information leakage between training and evaluation sets, leading to overly optimistic performance estimates. In addition, most approaches focus on discrete classification tasks and do not provide continuous safety metrics that capture temporal variability in driving behavior. Furthermore, the generalization capability of models across datasets and sensing modalities is rarely evaluated in a systematic manner.

These limitations highlight the need for a robust framework that supports leakage-free evaluation, continuous safety assessment, and cross-domain generalization. To address these challenges, the present work proposes a window-based deep learning framework with trip-level evaluation and a continuous safety scoring mechanism.

3. Methodology

3.1. Dataset

Experiments are conducted on the UAH-DriveSet benchmark [14], which contains 40 naturalistic driving trips recorded under six behavioral conditions: normal (D1), drowsy (D2), aggressive (D3), and three distracted variants (D4–D6). Each trip is captured using a smartphone mounted on the dashboard, providing GPS trajectory and inertial measurement unit (IMU) signals at 10 Hz. For this study, GPS-derived speed (km/h) and longitudinal acceleration (g) are used as the two input features, following the preprocessing pipeline of prior CAN-like simulation studies. Distracted driving conditions (D4–D6) are excluded, as their kinematic signatures are ambiguous and do not consistently correlate with the target unsafe behaviors (aggression and drowsiness). The final dataset comprises 28 trips: 14 labeled safe

(D1) and 14 labeled unsafe (D2 + D3). To assess generalizability to multi-channel CAN inputs, a synthetic dataset is additionally constructed. It contains 20 simulated trips (10 safe, 10 unsafe) generated using a physics-constrained mode, with speed bounded to [0,120] km/h and acceleration to [-0.5, 0.5] g for safe conditions, versus unbounded extremes for unsafe events. The synthetic set is used solely for held-out evaluation and is never seen during training.

3.2. Window-Based Signal Representation

Let

$$\mathbf{X} = \{\mathbf{x}_t \in \mathbb{R}^d \mid t = 1, \dots, N\} \quad (1)$$

denote the multivariate sensor time series for a single trip, where $d = 2$ (speed and longitudinal acceleration) and N is the trip length in samples. The signal is segmented into overlapping windows of fixed length $T = 200$ samples (20 s at 10 Hz) with step size $S = 20$ samples (2 s), giving window i as

$$\mathcal{W}_i = \{\mathbf{x}_t \mid t \in [i \cdot S, i \cdot S + T)\}. \quad (2)$$

Each window inherits the binary label of its parent trip ($y = 0$: safe, $y = 1$: unsafe). Segmentation yields approximately 3,400 windows across all 28 trips, with $\approx 46\%$ labeled unsafe.

3.3. Neural Network Architectures

1D-CNN. The input tensor of shape (200, 2) passes through three successive 1D-convolutional blocks. Block ℓ comprises:

- Conv1D with F_ℓ filters, kernel size 5, and same padding
- Batch Normalization followed by ReLU activation
- MaxPool1D with pool size 2

with filter counts $F_1 = 32$, $F_2 = 64$, $F_3 = 128$. The resulting feature map is flattened and passed through two fully connected layers ($256 \rightarrow 64$ units, ReLU), followed by a sigmoid output:

$$p_i = \sigma(\mathbf{w}^\top \mathbf{h} + b), \quad (3)$$

where $\mathbf{h} \in \mathbb{R}^{64}$ is the penultimate representation.

LSTM. The input tensor of shape (200, 2) is fed into a two-layer stacked LSTM with hidden sizes 128 and 64. Layer normalization is applied between LSTM layers. The final hidden state $\mathbf{h}_T \in \mathbb{R}^{64}$ is projected through a sigmoid layer to produce p_i (Eq. (3)).

3.4. Training Protocol

Both models are trained with binary cross-entropy loss weighted by inverse class frequency $w_c = N/(2N_c)$ to address label imbalance. The Adam optimizer is used with learning rate $\eta = 1 \times 10^{-3}$ and weight decay $\lambda = 1 \times 10^{-4}$, for up to 50 epochs with early stopping (patience = 7) monitored on validation AUC. Batch size is 64.

Table 1. Window-level performance on UAH-DriveSet (Trip-Level 5-Fold CV, mean $\pm$ std).

Model	Recall	F1	AUC	Accuracy
1D-CNN	0.913 $\pm$ 0.028	0.864 $\pm$ 0.030	0.941 $\pm$ 0.022	0.847 $\pm$ 0.032
LSTM	0.891 $\pm$ 0.033	0.867 $\pm$ 0.031	0.928 $\pm$ 0.027	0.831 $\pm$ 0.038
SVM	0.791 $\pm$ 0.049	0.782 $\pm$ 0.047	0.843 $\pm$ 0.039	0.764 $\pm$ 0.048
Random Forest	0.823 $\pm$ 0.043	0.813 $\pm$ 0.040	0.876 $\pm$ 0.034	0.798 $\pm$ 0.041

3.5. Evaluation Protocol

Trip-level 5-fold cross-validation is applied, ensuring that no window from the same trip appears in both training and validation splits simultaneously.

Window-level performance is reported using Accuracy, Precision, Recall, F1-Score, and AUC-ROC (mean $\pm$ std across folds). Trip-level classification uses the safety score threshold $\mathcal{S}_k = 0.5$.

3.6. Continuous Safety Score

To obtain a trip-level representation of driving safety, window-level predictions are aggregated into a continuous safety score. Let p_i denote the predicted probability of unsafe driving for the i -th window, and K be the total number of windows in a trip.

The safety score is defined as:

$$\mathcal{S}_k = \frac{\sum_{i=1}^K w_i p_i}{\sum_{i=1}^K w_i} \quad (4)$$

where w_i represents the importance weight of the i -th window. In this work, higher weights are assigned to windows exhibiting extreme acceleration patterns, reflecting their higher contribution to driving risk.

This formulation allows the safety score to emphasize critical driving events while maintaining robustness to noise.

4. Experimental Results

4.1. Window-Level Classification Performance

Table 1 reports window-level metrics across the 5-fold trip-level cross-validation. The 1D-CNN achieves the highest recall (0.913) and AUC (0.941), making it better suited for safety-critical applications where missed unsafe events carry greater cost than false alarms. Both deep learning models substantially outperform classical baselines (SVM and Random Forest with handcrafted statistical features: mean, standard deviation, skewness, kurtosis, and RMS per channel).

The recall advantage of 1D-CNN over SVM is +12.2 percentage points, confirming the value of automatic feature learning from raw CAN-like signals versus hand-crafted descriptors.

4.2. Trip-Level Safety Score Evaluation

Table 2 presents trip-level results obtained by thresholding the aggregated safety score $\mathcal{S}_k$ (Eq. 4) at 0.5. Despite the small number of available trips ($n = 28$), both models maintain high trip-level recall, confirming that unsafe trips consistently receive lower safety scores than safe trips.

Table 2. Trip-Level Classification Results (Safety Score Threshold = 0.5, $n = 28$ trips).

Model	Accuracy	Precision	Recall	F1
1D-CNN	0.821	0.812	0.96	0.834
LSTM	0.804	0.833	0.829	0.831

4.3. Zero-Shot Evaluation on Synthetic CAN Dataset

Table 3 reports results when both models are evaluated on the synthetic CAN dataset without retraining. The 1D-CNN achieves an AUC of 0.917, demonstrating robust transfer to simulated sensor inputs. Hard braking events (longitudinal acceleration < -0.3 g) and rapid acceleration episodes consistently receive low safety scores ($S_k < 0.3$), whereas steady-state segments score above 0.85.

Table 3. Zero-Shot Evaluation on Synthetic CAN Dataset (20 Simulated Trips, No Retraining).

Model	Accuracy	Recall	AUC
1D-CNN	0.883	0.900	0.917
LSTM	0.866	0.880	0.903

4.4. Per-Condition Recall Analysis

Table 4 breaks down window-level recall by behavioral condition for the 1D-CNN. Aggressive driving (D3) is detected with the highest recall (0.947), while drowsy driving (D2) is more challenging (0.879), consistent with its subtler kinematic footprint (gradual low-amplitude drift rather than sharp transients).

Table 4. 1D-CNN per-condition performance on UAH-DriveSet (F1 and AUC).

Condition	Label	F1	AUC
Normal D1	Safe	—	—
Aggressive D3	Unsafe	0.947	0.965
Drowsy D2	Unsafe	0.879	0.912

5. Discussion

The 1D-CNN recall advantage over LSTM (0.913 vs. 0.891) suggests that local short-term patterns—such as abrupt acceleration transients—are more discriminative than long-range temporal dependencies for the driving conditions studied here. This aligns with prior findings on CAN-based classification tasks [12, 9, 16]. The gap between aggressive and drowsy driving detection (recall 0.947 vs. 0.879) reveals a key challenge: drowsy driving is characterized by gradual, low-amplitude drift that may not produce strong local features within a 20-second window. Incorporating longer windows (e.g., 60 s) or explicit sequence modeling across consecutive windows is a natural extension.

A limitation of this study is the small number of trips in UAH-DriveSet ($n = 28$ after filtering), which constrains statistical power at the trip level. Future work should validate on larger naturalistic driving corpora (e.g., SHRP2) and explore integration with V2X communication systems for real-time safety score broadcast to roadside infrastructure—a direction directly relevant to cooperative intelligent transport systems (C-ITS).

6. Conclusion

This paper presented a window-based deep learning framework for driving safety assessment from CAN-like vehicle sensor data. Under trip-level cross-validation, the 1D-CNN achieved window-level recall of 0.913 and AUC of 0.941, outperforming LSTM and classical baselines (SVM, Random Forest) by more than 12 recall percentage points. A

continuous safety score derived from aggregated window-level probabilities correctly classified unsafe trips in over 82% of cases. Zero-shot evaluation on a synthetic CAN dataset confirmed generalizability to multi-source sensor inputs. These results support the deployment of data-driven safety monitoring in connected vehicle systems and smart city infrastructure.

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